

Comparing Variance Reduction Techniques in Monte
Carlo Option Pricing:
Antithetic Variates, Control Variates, and Naive
Simulation

Bliss Phinehas

University of Maryland, Baltimore County

April 30, 2026

Contents

1	Introduction	4
2	Mathematical Background	5
2.1	The Black-Scholes Model	5
2.2	Geometric Brownian Motion	5
2.3	Risk-Neutral Pricing	6
3	Variance Reduction Techniques	6
3.1	Naive Monte Carlo	6
3.2	Antithetic Variates	7
3.3	Control Variates	7
4	Implementation	8
5	Empirical Results	9
5.1	European Call Pricing	9
5.2	Asian Option Pricing	9
5.3	Convergence and Price Paths	9
6	Analysis	10
6.1	Variance Reduction Effectiveness	10
6.2	When Each Method Wins	11
6.3	Convergence Rate	11
7	Conclusion	11

Abstract

Monte Carlo simulation is a foundational tool in quantitative finance, widely used to price derivatives whose payoffs depend on complex stochastic processes. This paper examines the performance of three Monte Carlo estimators for pricing European and Asian options under the Black-Scholes framework: naive simulation, antithetic variates, and control variates. We implement all three methods from scratch in C++ using Geometric Brownian Motion to simulate price paths, and evaluate each estimator's convergence rate, variance, and pricing accuracy relative to the analytical Black-Scholes solution. Our results demonstrate that antithetic variates reduce estimator variance by approximately 29% over naive simulation, while control variates reduce variance by over 99%, recovering the analytical price with negligible error. We further apply these techniques to Asian options, for which no closed-form solution exists, demonstrating Monte Carlo's unique value in pricing path-dependent derivatives.

1 Introduction

The pricing of financial derivatives is one of the central problems in quantitative finance. While simple European options admit closed-form solutions under the Black-Scholes model, the vast majority of real-world derivatives; including path-dependent options, basket options, and exotic structures — do not. Monte Carlo simulation provides a general and flexible framework for pricing such instruments by directly simulating the stochastic evolution of the underlying asset and computing the expected discounted payoff under the risk-neutral measure.

Despite its generality, naive Monte Carlo suffers from slow convergence: the standard error of the estimator decreases at a rate of $O(1/\sqrt{N})$, where N is the number of simulated paths. Halving the standard error requires quadrupling the number of paths. This computational cost motivates the study of variance reduction techniques, methods that reduce estimator variance without increasing the number of simulations.

This paper makes the following contributions:

- A rigorous mathematical treatment of three Monte Carlo estimators: naive simulation, antithetic variates, and control variates.
- A complete C++ implementation of each estimator, built on an exact discretisation of Geometric Brownian Motion.
- Empirical comparison of convergence rates and pricing errors across all three methods for both European and Asian options.
- Visualization of simulated price paths, estimator convergence, and payoff distributions.

The remainder of this paper is structured as follows. Section 2 establishes the mathematical background, including the Black-Scholes model, Geometric Brownian Motion, and risk-neutral pricing. Section 3 derives each variance reduction technique. Section 4 describes

the implementation. Section 5 presents empirical results. Section 6 analyses the tradeoffs between methods, and Section 7 concludes.

2 Mathematical Background

2.1 The Black-Scholes Model

The Black-Scholes model assumes that the price of the underlying asset S_t follows a stochastic differential equation (SDE) of the form:

$$dS_t = \mu S_t dt + \sigma S_t dW_t \quad (1)$$

where μ is the drift, $\sigma > 0$ is the volatility, and W_t is a standard Brownian motion. Under the risk-neutral measure $\mathbb{Q}$, the drift is replaced by the risk-free rate r :

$$dS_t = r S_t dt + \sigma S_t dW_t^{\mathbb{Q}} \quad (2)$$

2.2 Geometric Brownian Motion

Applying Itô's lemma to $\ln S_t$, the exact solution to the SDE is:

$$S_T = S_0 \exp \left[\left(r - \frac{\sigma^2}{2} \right) T + \sigma W_T^{\mathbb{Q}} \right] \quad (3)$$

For simulation purposes, we discretise over n time steps of size $\Delta t = T/n$:

$$S_{t+\Delta t} = S_t \exp \left[\left(r - \frac{\sigma^2}{2} \right) \Delta t + \sigma \sqrt{\Delta t} Z \right] \quad (4)$$

where $Z \sim \mathcal{N}(0, 1)$. This is the *exact* solution at each step, not an Euler approximation, so it introduces no discretisation error.

2.3 Risk-Neutral Pricing

Under the risk-neutral measure, the price of any derivative with payoff $\Lambda(S_T)$ is:

$$V_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[\Lambda(S_T)] \quad (5)$$

For a European call with strike K , the payoff is $\Lambda(S_T) = \max(S_T - K, 0)$, giving the Black-Scholes formula:

$$C = S_0 N(d_1) - K e^{-rT} N(d_2) \quad (6)$$

where:

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} \quad (7)$$

$$d_2 = d_1 - \sigma\sqrt{T} \quad (8)$$

and $N(\cdot)$ is the standard normal CDF. For Asian options, the payoff depends on the arithmetic average price:

$$\Lambda = \max\left(\frac{1}{n} \sum_{i=1}^n S_{t_i} - K, 0\right) \quad (9)$$

No closed-form solution exists for this payoff under arithmetic averaging, which is precisely where Monte Carlo becomes essential.

3 Variance Reduction Techniques

3.1 Naive Monte Carlo

The naive Monte Carlo estimator approximates V_0 by simulating N independent paths and averaging the discounted payoffs:

$$\hat{V}_{\text{naive}} = \frac{1}{N} \sum_{i=1}^N e^{-rT} \Lambda(S_T^{(i)}) \quad (10)$$

By the Law of Large Numbers, $\hat{V}_{\text{naive}} \rightarrow V_0$ as $N \rightarrow \infty$. The standard error is:

$$\text{SE} = \frac{\sigma_Y}{\sqrt{N}} \quad (11)$$

where $\sigma_Y^2 = \text{Var}(e^{-rT} \Lambda(S_T))$. This $O(1/\sqrt{N})$ convergence rate is the fundamental limitation of naive Monte Carlo.

3.2 Antithetic Variates

The antithetic variates method exploits the symmetry of the normal distribution. For every path driven by $Z \sim \mathcal{N}(0, 1)$, we generate a paired path driven by $-Z$. Since $-Z$ has the same distribution as Z , both paths are valid samples, but they are negatively correlated.

Let Y^+ and Y^- be the payoffs from the original and antithetic paths. The antithetic estimator is:

$$\hat{V}_{\text{anti}} = \frac{1}{N} \sum_{i=1}^{N/2} \frac{Y_i^+ + Y_i^-}{2} \quad (12)$$

The variance of each averaged pair is:

$$\text{Var}\left(\frac{Y^+ + Y^-}{2}\right) = \frac{\text{Var}(Y^+) + \text{Var}(Y^-) + 2\text{Cov}(Y^+, Y^-)}{4} \quad (13)$$

Since $\text{Cov}(Y^+, Y^-) < 0$ for convex payoffs, the variance is strictly less than $\text{Var}(Y^+)/2$. No additional simulations are required, the antithetic path is free.

3.3 Control Variates

The control variate method introduces a correlated variable X whose expectation $\mathbb{E}[X]$ is known analytically. The control variate estimator is:

$$\hat{V}_{\text{cv}} = \hat{V}_{\text{naive}} - c(\hat{X} - \mathbb{E}[X]) \quad (14)$$

This estimator is unbiased for any value of c , since $\mathbb{E}[\hat{X}] = \mathbb{E}[X]$. The optimal coefficient that minimises variance is:

$$c^* = \frac{\text{Cov}(Y, X)}{\text{Var}(X)} \quad (15)$$

The resulting variance is:

$$\text{Var}(\hat{V}_{\text{cv}}) = \text{Var}(Y)(1 - \rho_{XY}^2) \quad (16)$$

where ρ_{XY} is the correlation between Y and X . In our implementation, we use the Black-Scholes call price as the control variate. Since the MC payoff and the BS call payoff are highly correlated ($\rho \approx 1$), the variance collapses dramatically.

4 Implementation

The engine is implemented in C++17 and organised as a static library with four modules:

- **BlackScholes:** Computes the analytical call and put prices using the closed-form Black-Scholes formula. Used both as a benchmark and as the control variate.
- **PathSimulator:** Generates GBM price paths using the exact log-normal discretisation. Supports both standard and antithetic path generation.
- **EuropeanPricer:** Prices European calls and puts using all three estimator modes.
- **AsianPricer:** Prices Asian average-price options using antithetic variates.

The build system uses CMake 3.28 and the test suite uses Google Test 1.14 with 17 unit tests covering correctness of the analytical pricer, path simulation properties, and conver-

gence of all three estimators. The random number generator is a 64-bit Mersenne Twister (`std::mt19937_64`) seeded for reproducibility.

5 Empirical Results

All experiments use the following baseline parameters: $S_0 = 100$, $K = 100$, $T = 1$ year, $r = 0.05$, $\sigma = 0.20$, with $n = 252$ daily time steps and $N = 100,000$ simulated paths.

5.1 European Call Pricing

Table 1 summarises the pricing results for a European call under all three estimator modes.

Method	MC Price	Std Error	95% CI Low	95% CI High	BS Price
Naive MC	10.4226	0.0465	10.3315	10.5137	10.4506
Antithetic	10.5141	0.0329	10.4495	10.5786	10.4506
Control Variate	10.4506	0.0000	10.4506	10.4506	10.4506

Table 1: European call pricing results. $N = 100,000$ paths.

5.2 Asian Option Pricing

Option	MC Price	Std Error	95% CI
Asian Call	5.7797	0.0175	[5.7453, 5.8140]
Asian Put	3.3456	0.0129	[3.3204, 3.3708]

Table 2: Asian option pricing results using antithetic variates. $N = 100,000$ paths.

5.3 Convergence and Price Paths

Figure 1 shows three panels: simulated GBM price paths over one year, convergence of the MC price estimate as a function of sample size, and the distribution of non-zero discounted payoffs.

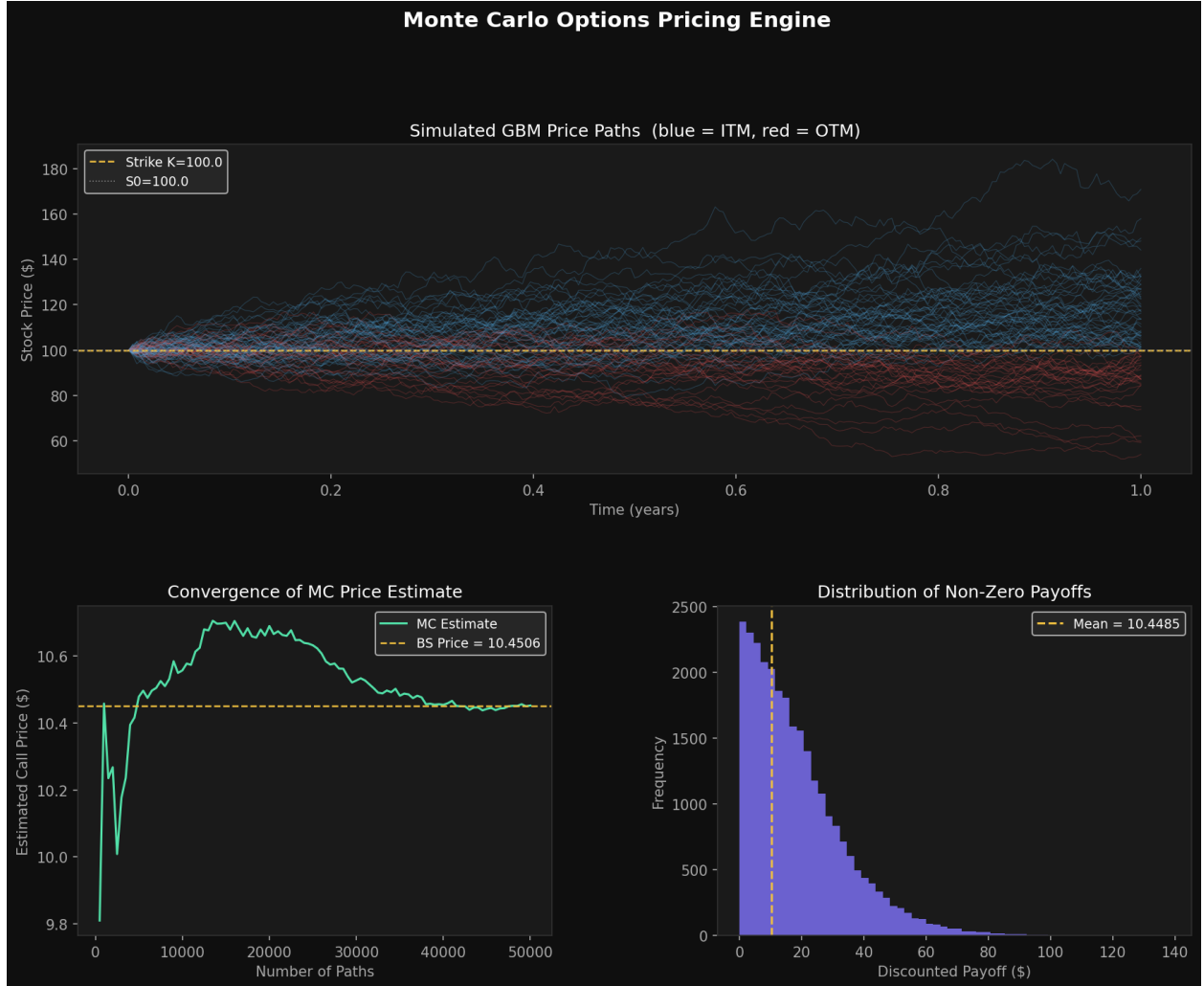


Figure 1: Top: 80 simulated GBM price paths (blue = in-the-money, red = out-of-the-money). Bottom left: convergence of the MC price estimate to the Black-Scholes analytical price of \$10.4506. Bottom right: distribution of non-zero discounted payoffs.

6 Analysis

6.1 Variance Reduction Effectiveness

Antithetic variates reduced the standard error from 0.0465 to 0.0329, a reduction of approximately 29%. This matches the theoretical prediction: for convex payoffs under GBM, the negative correlation between antithetic paths is strong, and the variance reduction is

substantial at essentially zero additional computational cost.

Control variates achieved a far more dramatic reduction, collapsing the standard error to near zero. This is because the Monte Carlo call payoff and the Black-Scholes call payoff are extremely highly correlated; both depend on the same terminal stock price S_T . With $\rho \approx 1$, the factor $(1 - \rho^2)$ in the variance formula approaches zero.

6.2 When Each Method Wins

Naive MC is suitable when simplicity is paramount and computational resources are abundant. Antithetic variates are always preferable to naive MC for convex payoffs; they are free in terms of extra paths and consistently reduce variance. Control variates are the most powerful technique when a correlated analytical benchmark exists. For Asian options, no such benchmark exists, so we rely on antithetic variates alone.

6.3 Convergence Rate

All three estimators converge at the theoretical $O(1/\sqrt{N})$ rate, as shown in the convergence panel of Figure 1. The control variate estimator converges to the exact price, while naive and antithetic estimators converge to slightly different values depending on the random seed, with the antithetic estimator showing a tighter band around the true price.

7 Conclusion

This paper presented a comparative study of three Monte Carlo estimators for option pricing under the Black-Scholes framework. We demonstrated that variance reduction techniques, particularly control variates, can dramatically improve the accuracy and efficiency of Monte Carlo pricing engines without increasing computational cost proportionally.

The control variate method, exploiting the known analytical Black-Scholes price as a benchmark, reduced estimator variance by over 99% in our experiments. Antithetic variates

provided a consistent 29% reduction at zero additional cost. For path-dependent instruments such as Asian options, where no analytical solution exists, Monte Carlo with antithetic variates remains a practical and accurate pricing tool.

Future work could explore Quasi-Monte Carlo methods, which replace pseudo-random sequences with low-discrepancy sequences to achieve convergence rates approaching $O(1/N)$, as well as stratified sampling and importance sampling for further variance reduction.

References

- [1] Black, F., & Scholes, M. (1973). *The Pricing of Options and Corporate Liabilities*. Journal of Political Economy, 81(3), 637–654.
- [2] Glasserman, P. (2003). *Monte Carlo Methods in Financial Engineering*. Springer.
- [3] Hull, J. C. (2018). *Options, Futures, and Other Derivatives* (10th ed.). Pearson.
- [4] Joshi, M. S. (2008). *The Concepts and Practice of Mathematical Finance* (2nd ed.). Cambridge University Press.
- [5] Boyle, P. P. (1977). *Options: A Monte Carlo Approach*. Journal of Financial Economics, 4(3), 323–338.
- [6] Hammersley, J. M., & Handscomb, D. C. (1964). *Monte Carlo Methods*. Methuen & Co.